The Interplay of AI-and-RAN: Dynamic Resource Allocation for Converged 6G Platform

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1 Bibliographic Information

- **Title**: The Interplay of AI-and-RAN: Dynamic Resource Allocation for Converged 6G Platform
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- **Source**: IEEE INFOCOM 2025 Workshop (arXiv:2503.07420)
- **Keywords**: O-RAN, 5G, AI, xApp, RIC, SAC

1.1 One-line summary

A study targeting the **AI-and-RAN** scenario defined by the AI-RAN Alliance (where AI workloads and RAN workloads share a single accelerated computing infrastructure), proposing a convergence architecture **CAORA** that extends O-RAN's NRT-RIC, and solving the problem of distributing GPU resources (MIG) between two workloads using **SAC (Soft Actor-Critic) reinforcement learning**. In simulations, 100% RAN infrastructure resource utilization was achieved while satisfying approximately 99% of RAN workload requests.

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2 Problem Awareness (Motivation)

Existing RAN infrastructure **dedicatedly allocates fixed computing resources for network functions**. This static operation creates two inefficiencies simultaneously.

1. **Off-peak resource waste**: Accelerated computing resources remain idle during periods of low RAN load.
2. **AI workload incapacity**: The same infrastructure cannot handle the rapidly increasing demand for GenAI/edge AI inference simultaneously.

The essence of the problem is that the nature of the two workloads is opposite.

Characteristics	RAN workload	AI workload
Time Constraint	latency-sensitive, real-time (non-elastic)	relatively flexible (elastic)
Resource Demand	Relatively predictable depending on carrier settings	Large variability by use case
Priority	High (Protocol stack operation required)	Best-effort within available resources

Therefore, the starting point of this paper is the need for a mechanism that tracks demand in real-time and dynamically updates policies, instead of static allocation. The authors point out that existing research has addressed only one side—either **RAN optimization** or **AI workload scheduling**—and has failed to present an architecture and optimization formulation for the **coexistence** of the two.

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3 CAORA Architecture

CAORA (Converged AI-and-ORAN Architecture) is a 4-layer hierarchical control architecture based on the O-RAN specification.

3.1 Hierarchy Configuration

- **Layer 1 — External Cloud**: The layer where AI/ML workloads reside. It houses edge AI inferencing, generative AI, robotics applications, etc., and connects to the RAN infrastructure (compute cluster) via the **AI Deployment API**.
- **Layer 2 — Programmable RAN**: O-RU / O-DU / O-CU according to O-RAN disaggregation.
- **Layer 3 — Controller (NRT-RIC)**: It is equipped with custom xApps responsible for traffic prediction, resource allocation, and monitoring. It communicates with RAN components via the **E2 interface** and makes rapid resource decisions.
- **Layer 4 — Resource Management (E2E Orchestrator)**: The entity that manages the actual compute cluster (GPU, **MIG**, NIC). It contains an SAC agent.

3.2 Core data flow: Y1 interface

The structural core of this paper is the **Y1 interface**.

- **Monitoring xApp** (Layer 3) continuously collects RAN KPIs such as bandwidth utilization, system latency, and network load.
- This radio analytics is exposed to the E2E orchestrator (Layer 4) through the **Y1 interface**.
- The E2E orchestrator receives information as a **Y1 consumer** and determines resource allocation using SAC.

That is, the structure is *"xApp = sensor, E2E orchestrator (SAC) = brain"*, and the SAC agent supports coexistence through three mechanisms: (i) **predictive resource planning (predictive analysis)** utilizing past patterns, (ii) **real-time adaptation** based on current conditions, and (iii) **priority management** prioritizing the RAN during peak times.

This architecture transforms traditional RAN infrastructure into a **multi-purpose system** that supports both network operations and AI workloads.

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4 System Model

4.1 Task expression

Let the entire task set be K and the individual task be $k \in K$:

$$k = \{\tau, r, p\}$$

- τ: task type (RAN / AI)
- $r = [r_i,\ \forall i = 1, 2, \ldots, N]$: Resource Requirement Vector (Requirement by resource type)
- $p \in [0, 1]$: Priority (Real-time signal processing > Non-real-time analysis/reporting)

Since resource demand changes over time, it is denoted as $r(k, t)$ and $p(k)$.

4.2 Resource Demand Definition

The resource demand at time t of task k is defined as the L_1 norm of the demand vector.

$$\Delta(k, t) = \|r(k, t)\|_1 = \sum_{i=1}^{n} r_i(k, t)$$

The total system demand is the sum of all tasks.

$$\Delta_{\text{total}}(t) = \sum_{k \in K} \Delta(k, t)$$

4.3 System Dynamics

The entire workload $W(t)$ evolves through a balance of task **creation**, **completion**, and **resource contention**.

$$W(t) = \int_0^t \left(\sum_{k \in K} P(k, t)\, \Delta(k, t) \ - \ \sum_{k \in K} P_c(k, t)\, \Delta(k, t)\, C(k, t) \right) dt$$

- $P(k, t)$: task generation probability
- $P_c(k, t) = \alpha \cdot p(k)$: task completion probability (α is system efficiency coefficient, $p(k)$ is priority)

- $C(k,t)$: Resource contention for task k

The first term represents new incoming load, and the second term represents load that is processed and discharged faster as its priority (and efficiency) increases.

4.4 Problem formulation

It boils down to a dynamic allocation problem of maximizing priority-reflecting utility while ensuring that the sum of RAN and AI allocations within a normalized shared resource pool does not exceed available resources. The key constraints are of the following form.

$$r_{\mathrm{RAN}}(t) + r_{\mathrm{AI}}(t) \;\leq\; \mathcal{R}_{\max}$$

where $\mathcal{R}_{\max}$ is the total amount of available computing resources, which is modeled based on NVIDIA MIG in this paper (see Evaluation section below).

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5 SAC-based dynamic resource adaptation

The E2E orchestrator uses the **SAC (Soft Actor-Critic)** algorithm, which is suitable for continuous action spaces. SAC is suitable for adjusting continuous resource allocations by stably maintaining the search-exploitation balance through entropy normalization.

The components of are organized as follows.

- **State**: Network status indicators such as current RAN/AI resource demand and previous allocation history delivered by the monitoring xApp (based on bandwidth utilization, system latency, network load).
- **Action**: Adjustment (increase/decrease) of resource quotas for RAN and AI.
- **Reward**: Rewards the degree of task completion weighted by priority, while simultaneously reflecting **utilization** to ensure that infrastructure resources do not remain idle. A policy is learned to prioritize the allocation of resources to the RAN during peak times through time-varying weights.

This design strikes a balance between the two goals of *"RAN real-time guarantee"* and *"best-effort accommodation for AI workloads"* within a single compensation function.

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6 Performance Evaluation Results

6.1 Experimental Environment

- Pure **Python simulation** (O-RAN based assumption).
- Resources are modeled in **NVIDIA MIG (Multi-Instance GPU)** units. In other words, resource pools and demand are represented not by abstract "resources," but by the number of instances created by partitioning a physical GPU.
- Evaluate policies in a dynamic scenario where RAN/AI demand fluctuates over time.

6.2 Key Results

- Meets **approx. 99%** of RAN workload requests — Reliably ensures RAN for real-time demands.
- Achieved **100%** utilization of RAN infrastructure resources — AI workload absorbs off-peak idle resources.
- When RAN demand surges during the peak period, SAC immediately reduces AI allocation and reallocates it to RAN → the operation of priority-weighted reward is demonstrated.

- AI workloads are accommodated at a lower priority level than RAN, but are continuously processed within the limits of available resources.

In conclusion, it was shown that dynamic resource sharing maximizes infrastructure utility compared to static allocation without compromising RAN performance requirements.

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7 Limit

The limitations revealed by the paper itself (or structurally) are as follows, and these serve as the entry point for subsequent research.

1. **Pure Simulation Verification**: No actual RAN stack (L1/MAC), user count abstracted as MIG demand. No measured traffic dataset used.
2. **Resource-level coarseness**: Treat tasks as MIG units → Fractional/variable allocation or wireless resource (PRB/TB) level control is out of the scope.
3. **NRT-RIC Time Scale**: Based on Y1/E2, the control cycle is in the **10 ms to second range**. In other words, it is "macro-level computing resource pool allocation," not sub-TTI scheduling.
4. **Blackboxing AI workload**: Modeling the AI task based solely on the "number of required MIGs" without examining its internal structure (token, attention, etc.).

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8 Research Gap and Thesis Differentiation Axis

This paper is situated on the **AI-and-RAN (computing coexistence/resource sharing)** axis among the three branches of AI-RAN. On the other hand, my research direction, **importance-aware MAC scheduling**, is closer to **AI-for-RAN (intelligent control)**, and the two studies do not compete but rather deal with **different layers and granularities**.

8.1 Differentiation Comparison

Category	Main Paper (CAORA)	Thesis Direction
Control Principal	NRT-RIC xApp + External E2E orchestrator	GPU-collocated **L2 MAC scheduler**
Timescale	NRT-RIC (10 ms~sec), via Y1/E2	**sub-TTI**, direct connection within node
Resource Unit	MIG (GPU Batch)	**PRB / TB** (Wireless Resource)
AI workload handling	Black box (number of MIGs required)	**token importance**(entropy · attention) as scheduler utility input
Meaning of "AI-and-RAN"	RAN+AI **compute coexistence** on GPU	RAN **importance-aware** scheduling offloaded VLM traffic
Evaluation Metrics	Request Fulfillment Rate, Resource Utilization Rate	**end-to-end task utility per resource**
Verification Stack	Python Simulation	Sionna → cuMAC on GH200, ROS2 Traffic

8.2 Key Differentiation Argument

- **Difference in meaning of AI workload**: In this paper, AI workload refers to *computational load on the GPU*. On the other hand, in my use case, multi-robot VLM offloading is *uplink traffic itself*.

Therefore, while this paper addresses "GPU sharing of AI computation and RAN computation," I address a problem one step closer: "RAN scheduling traffic generated by AI to wireless resources according to the importance of its content."

- **Timescale Differentiation**: Since this paper accurately demonstrates the macroscopic resource allocation of the NRT-RIC/Y1 path, the argument that sub-TTI control cannot be externalized to this path (the inevitability of GPU-collocated MAC control) serves as a direct contrast to this paper.
- **Room for Expansion of Prediction Elements**: While this paper discusses predictions utilizing the historical patterns of the monitoring xApp, it does not address the approach of **proactively pre-allocating future PRBs by leveraging channel quality and traffic predictions based on token content**. This is the core area of my contribution.

8.3 Positioning Memo

- As this paper is one of the most direct prior studies in the field of predictive AI-and-RAN orchestration, it is cited as a **convergence backbone** in seminars and as an **upper limit for macro resource allocation** in dissertations.
- By pairing this paper with the sequel (arXiv:2507.09124, *Proactive AI-and-RAN Workload Orchestration*) and comparing how the reactive/predictive elements of this paper are extended to proactive, the gap can be anchored more clearly.

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9 View at a glance

This paper is a representative macro resource allocation study of AI-and-RAN that solves the question of "how RAN and AI will share GPUs (MIG) on accelerated computing nodes" using NRT-RIC + SAC. My contribution lies *inside*: **importance-aware scheduling of traffic accepted by the RAN (especially VLM offloading) in sub-TTI MACs based on token importance** — a micro-control area left vacant in this paper.